Supplement to "Silent Aliasing and Certified Sampling Design for Fixed Fourier-Feature Models": Full Proofs and Additional Results

Weiping Yan

Theorem map (main ↔ supplement). Theorem 1 (structured indistinguishability) is proved in §1; Theorem 2 (off-grid randomization: jitter law + i.i.d. concentration) in §3; Proposition 1 (the continuum certificate) in §6 (Prop. S4). Supplement items: Lemma S1 (LS stability), Prop. S1 (Riesz conversion + exact error decomposition), Prop. S2 (average-case, background), Prop. S3 (surrogate controls aliasability), Prop. S4 (certificate). The detection dichotomy (§4) is standard and demoted from the main line.

Setting and notation as in the main paper: $e_\omega(x) = e^{i2\pi\omega x}$ on $[0,1)^d$, model set $\Lambda = \{\omega_1, \ldots, \omega_m\}$, samples $T = \{t_j\}_{j=1}^N$, synthesis matrix $\boldsymbol{\Phi}_{jk} = e_{\omega_k}(t_j)$, sample atom $\boldsymbol{\phi}_\nu = (e^{i2\pi\nu t_j})_j$, visibility $v_T(\nu) = \|(\boldsymbol{I} - \boldsymbol{P}_\Lambda)\boldsymbol{\phi}_\nu\|/\sqrt{N}$, coefficient aliasability $a_T(\nu) = \|\boldsymbol{\Phi}^\dagger\boldsymbol{\phi}_\nu\|$, function-space aliasability $a_{L^2,T}(\nu) = (\Delta c^* G_{L^2} \Delta c)^{1/2}$ with $\Delta c = \boldsymbol{\Phi}^\dagger\boldsymbol{\phi}_\nu$. Every closed form is also checked against independent simulation in `tests/`.

Lemma S 1 (LS stability; standard)**.** *If $\sigma_{\min}(\boldsymbol{\Phi}) > 0$ and $f_{\text{out}} = 0$, then $\|\hat{\boldsymbol{c}} - \boldsymbol{c}^\star\| \le \|\boldsymbol{\varepsilon}\|/\sigma_{\min}(\boldsymbol{\Phi})$, and for white noise $\mathbb{E}\|\hat{\boldsymbol{c}} - \boldsymbol{c}^\star\|^2 = \sigma^2 \operatorname{tr}((\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1})$.*

Proof. $\hat{\boldsymbol{c}} - \boldsymbol{c}^\star = \boldsymbol{\Phi}^\dagger\boldsymbol{\varepsilon}$ and $\|\boldsymbol{\Phi}^\dagger\| = 1/\sigma_{\min}(\boldsymbol{\Phi})$; the trace identity is $\mathbb{E}\|\boldsymbol{\Phi}^\dagger\boldsymbol{\varepsilon}\|^2 = \sigma^2 \operatorname{tr}(\boldsymbol{\Phi}^\dagger\boldsymbol{\Phi}^{\dagger *}) = \sigma^2 \operatorname{tr}((\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1})$. □

1 Proof of Theorem 1: structured indistinguishability

(i) $v_T(\nu) = 0$ iff $(\boldsymbol{I} - \boldsymbol{P}_\Lambda)\boldsymbol{\phi}_\nu = 0$ iff $\boldsymbol{\phi}_\nu \in \operatorname{range}\boldsymbol{\Phi}$, which is precisely the statement that some coefficient vector $\boldsymbol{c}$ has $\boldsymbol{\Phi}\boldsymbol{c} = \boldsymbol{\phi}_\nu$, i.e. the unit tone at ν agrees with a model element at every sample. For the grid statement, let $t_j = g_j/Q$ with $g_j \in \{0, \ldots, Q-1\}$ and $\nu = \omega_k + rQ$, $r \in \mathbb{Z}$. Then

$$e^{i2\pi\nu g_j/Q} = e^{i2\pi\omega_k g_j/Q}\, e^{i2\pi r g_j} = e^{i2\pi\omega_k g_j/Q},$$

since $rg_j \in \mathbb{Z}$; hence $\boldsymbol{\phi}_\nu = \boldsymbol{\phi}_{\omega_k}$ entrywise, for *every* subset of the grid. Consequently $v_T(\nu) = 0$; moreover, because $\sigma_{\min}(\boldsymbol{\Phi}) > 0$ makes the LS solution unique, the fit of $y = \boldsymbol{\phi}_\nu$ is the unique $\boldsymbol{c}$ with $\boldsymbol{\Phi}\boldsymbol{c} = \boldsymbol{\phi}_{\omega_k}$, namely the k-th unit vector, and the residual is exactly zero.

(ii) By (i), $S_T f_1 = S_T f_2$ pointwise. For any noise mechanism that acts on $S_T f$ (additive or otherwise, as long as its law depends on the signal only through $S_T f$), the observation distributions under f_1 and f_2 are therefore identical. Let $\hat{f} = \delta(\boldsymbol{y})$ be any estimator. For every realization $\boldsymbol{y}$,

$$\|\delta(\boldsymbol{y}) - f_1\|_{L^2} + \|\delta(\boldsymbol{y}) - f_2\|_{L^2} \;\ge\; \|f_1 - f_2\|_{L^2} = |a|\, \|e_\nu - e_{\omega_k}\|_{L^2} = |a|\sqrt{2},$$

using the triangle inequality and, for the last equality, that e_ν and e_{ω_k} are distinct integer frequencies, hence orthonormal in $L^2[0,1)$. Thus $\max_i \|\delta(\boldsymbol{y}) - f_i\| \ge |a|/\sqrt{2}$ pointwise (a two-point bound in the sense of Le Cam, here even realization-wise).

(iii) If $[\boldsymbol{\Phi}\ \boldsymbol{\phi}_\nu]$ has full column rank then $\boldsymbol{\phi}_\nu \notin \operatorname{range}\boldsymbol{\Phi}$, so $v_T(\nu) > 0$ by (i). □

Remark 1. *The full-column-rank hypothesis on $\boldsymbol{\Phi}$ requires $N \ge m$ and the elements of Λ to be distinct modulo Q (two atoms congruent mod Q have identical columns on any grid subset). "Any N" claims are always to be read as "any grid-subset size N for which this rank condition holds".*

2 Prop. S1: Riesz conversion and error decomposition

Proposition S 1 (Riesz conversion and exact decomposition)**.** *With $G_{L^2}[k,\ell] = \langle e_{\omega_\ell}, e_{\omega_k}\rangle_{L^2}$ and extreme eigenvalues $\lambda_{\min}, \lambda_{\max} > 0$ (distinct frequencies), $\lambda_{\min}\|\boldsymbol{c}\|^2 \le \|\sum_k c_k e_{\omega_k}\|^2_{L^2} \le \lambda_{\max}\|\boldsymbol{c}\|^2$, and $\|\hat f - f\|^2_{L^2} = \Delta\boldsymbol{c}^* G_{\Lambda\Lambda}\Delta\boldsymbol{c} - 2\,\mathrm{Re}\,\Delta\boldsymbol{c}^* G_{\Lambda,\mathrm{out}}\boldsymbol{a} + \boldsymbol{a}^* G_{\mathrm{out}}\boldsymbol{a}$ (modeled-component, cross, truncation).*

Proof. **(i)** G_{L^2} is the Gram matrix of $\{e_{\omega_k}\}$ in $L^2[0,1)$: $\|\sum_k c_k e_{\omega_k}\|^2_{L^2} = \boldsymbol{c}^* G_{L^2}\boldsymbol{c}$, and the Rayleigh quotient of the Hermitian PSD matrix G_{L^2} lies in $[\lambda_{\min}, \lambda_{\max}]$. The entries are $\int_0^1 e^{i2\pi(\omega_\ell - \omega_k)t}\,dt = \frac{e^{i2\pi\delta}-1}{i2\pi\delta} = e^{i\pi\delta}\,\frac{\sin\pi\delta}{\pi\delta}$ for $\delta = \omega_\ell - \omega_k \neq 0$, and 1 on the diagonal. G_{L^2} is singular iff the atoms are linearly dependent in L^2, which for finitely many distinct frequencies never happens (distinct exponentials are linearly independent); hence $\lambda_{\min} > 0$ for distinct frequencies.

(ii) Expand $\hat f - f = \sum_k \Delta c_k e_{\omega_k} - \sum_l a_l e_{\nu_l}$ and evaluate $\|\hat f - f\|^2_{L^2}$ as the quadratic form of the joint Gram matrix of $(\{\omega_k\}, \{\nu_l\})$ applied to $(\Delta\boldsymbol{c}, -\boldsymbol{a})$; grouping blocks gives the stated decomposition. For all-integer frequencies the off-diagonal blocks vanish (orthonormality), killing the cross term. $\square$

3 Proof of Theorem 2: off-grid randomization

We use the following complex Hoeffding bound throughout: if $Z_1, \dots, Z_N$ are independent with $|Z_j| \le 1$, then for $\bar Z = \frac{1}{N}\sum_j Z_j$,

$$\Pr\left(|\bar Z - \mathbb{E}\bar Z| \ge \varepsilon\right) \ \le\ 4\exp(-N\varepsilon^2/4), \tag{1}$$

obtained by applying the real Hoeffding inequality to the real and imaginary parts (each in $[-1,1]$) with radius $\varepsilon/\sqrt{2}$ and a union bound.

3.1 Part (a): jitter

Let $t_j = g_j/Q + \eta_j$ with g_j grid points and η_j i.i.d. mean-zero with characteristic function χ_η, and $\nu - \omega_k = rQ$, $r \in \mathbb{Z}\setminus\{0\}$. The fold coherence is

$$C = \frac{1}{N}\sum_j e^{i2\pi(\nu-\omega_k)t_j} = \frac{1}{N}\sum_j \underbrace{e^{i2\pi r g_j}}_{=1} e^{i2\pi rQ\eta_j} = \frac{1}{N}\sum_j e^{i2\pi rQ\eta_j},$$

an empirical characteristic function with $\mathbb{E}C = \chi_\eta(2\pi rQ)$ and concentration (1): $|C - \chi_\eta(2\pi rQ)| \le 2\sqrt{\log(4/\delta)/N}$ with probability $\ge 1-\delta$. For Gaussian jitter, $\chi_\eta(u) = e^{-u^2\sigma_t^2/2}$, so $\chi_\eta(2\pi rQ) = e^{-2(\pi rQ\sigma_t)^2}$; for uniform jitter of width w, $\chi_\eta(2\pi rQ) = \mathrm{sinc}(rQw)$.

From coherence to visibility. Let $\boldsymbol{b} = \boldsymbol{\Phi}^*\boldsymbol{\phi}_\nu/N \in \mathbb{C}^m$ and $\boldsymbol{G}_N = \boldsymbol{\Phi}^*\boldsymbol{\Phi}/N$. Then

$$v_T(\nu)^2 = \frac{1}{N}\|\boldsymbol{\phi}_\nu\|^2 - \frac{1}{N}\boldsymbol{\phi}_\nu^*\boldsymbol{P}_\Lambda\boldsymbol{\phi}_\nu = 1 - \boldsymbol{b}^*\boldsymbol{G}_N^{-1}\boldsymbol{b}.$$

Assume additionally—as in the theorem statement—that the grid points g_j are drawn i.i.d. uniformly from the grid (a uniformly random subset drawn without replacement obeys the same bounds with the same constants by Serfling's inequality for sampling without replacement) and that the elements of Λ are distinct modulo Q (otherwise two Gram columns coincide on every grid subset and $\boldsymbol{G}_\infty \neq \boldsymbol{I}$, breaking the perturbation step below). Each entry of $\boldsymbol{b}$ other than the fold entry, and each off-diagonal entry of $\boldsymbol{G}_N$, is an empirical mean of independent unit-modulus terms whose expectation vanishes (distinct residues modulo Q average to zero over the uniform grid, and the jitter factor only multiplies by a number of modulus ≤ 1); the fold entry of $\boldsymbol{b}$ is C above. By (1) and a union bound over these $m + m^2$ quantities, with probability $\ge 1-\delta$ all of them are within $\varepsilon = 2\sqrt{\log(4(m+m^2)/\delta)/N}$ of their means. Writing $\boldsymbol{G}_N = \boldsymbol{I} + \boldsymbol{E}$ with $\|\boldsymbol{E}\| \le m\varepsilon$ and $\boldsymbol{b} = C\boldsymbol{e}_k + \boldsymbol{u}$ with $\|\boldsymbol{u}\| \le \sqrt{m}\,\varepsilon$, a first-order expansion of $\boldsymbol{b}^*\boldsymbol{G}_N^{-1}\boldsymbol{b}$ gives

$$\left|v_T(\nu)^2 - \left(1 - |\chi_\eta(2\pi rQ)|^2\right)\right| \ \le\ c\,m\,\varepsilon$$

for an absolute constant c (valid while $m\varepsilon < 1/2$), where $A := 1 - |\chi_\eta(2\pi rQ)|^2 \in [0,1]$. This is a bound on the *squared* visibility; it does not transfer to v_T by a first-order expansion when $A \to 0$. Two correct finite-sample statements follow. *(Uniform.)* Since $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$ for $x, y \ge 0$,

$$\left|v_T(\nu) - \sqrt{A}\right| \ \le\ \sqrt{c\,m\,\varepsilon} \qquad \text{always,}$$

so the visibility is within $O\big((m\varepsilon)^{1/2}\big) = O\big(m^{1/2}(\log/N)^{1/4}\big)$ of $\sqrt{A}$. *(Refined, A bounded below.)* If $A \geq A_0 > 0$ then, since $\sqrt{\cdot}$ is $\frac{1}{2\sqrt{A_0}}$-Lipschitz on $[A_0, \infty)$, $\big|v_T(\nu) - \sqrt{A}\big| \leq c\, m\, \varepsilon/(2\sqrt{A_0}) = O(m\varepsilon/\sqrt{A})$. *Small-jitter linear law.* For Gaussian jitter, $\sqrt{A} = \sqrt{1 - e^{-4(\pi r Q \sigma_t)^2}} = 2\pi|r|Q\sigma_t\,(1 + o(1))$ as $\sigma_t \to 0$ (from $1 - e^{-2u} = 2u + O(u^2)$, $u = 2(\pi r Q \sigma_t)^2$). The empirical linear law $v_T(\nu) \approx 2\pi|r|Q\sigma_t$ therefore holds in the *joint* regime where the deterministic signal dominates the finite-N fluctuation, $2\pi|r|Q\sigma_t \gg \sqrt{c\, m\, \varepsilon}$, i.e. $\sigma_t \gg (m\varepsilon)^{1/2}/(2\pi|r|Q)$ with $\varepsilon = 2\sqrt{\log(4(m + m^2)/\delta)/N}$ (equivalently, N large enough for the fixed jitter to be resolved); otherwise the $O((m\varepsilon)^{1/2})$ fluctuation dominates.

3.2 Part (b): i.i.d. samples

Entries of $\boldsymbol{G}_N - \boldsymbol{G}_\infty$ and of $\boldsymbol{b}_\nu = \boldsymbol{\Phi}^* \boldsymbol{\phi}_\nu / N$ (for every $\nu \in \Omega$) are empirical means of independent unit-modulus variables; by (1) and a union bound over the $m^2 + mK$ of them, with probability $\geq 1 - \delta$ every such quantity is within $\varepsilon = 2\sqrt{\log(4(mK + m^2)/\delta)/N}$ of its limit. Under the hypotheses, the limiting coherences vanish, so $\|\boldsymbol{b}_\nu\| \leq \sqrt{m}\,\varepsilon$ for all $\nu \in \Omega$, and $\|\boldsymbol{G}_N - \boldsymbol{G}_\infty\| \leq \|\cdot\|_F \leq m\varepsilon$, hence $\lambda_{\min}(\boldsymbol{G}_N) \geq \lambda_{\min} - m\varepsilon > 0$. Therefore, uniformly over Ω,

$$a_T(\nu) = \|\boldsymbol{G}_N^{-1} \boldsymbol{b}_\nu\| \leq \frac{\|\boldsymbol{b}_\nu\|}{\lambda_{\min}(\boldsymbol{G}_N)} \leq \frac{\sqrt{m}\,\varepsilon}{\lambda_{\min} - m\varepsilon}. \qquad \square$$

Remark 2. *The bound is a union bound and is loose by design (Fig. 1b of the main paper); its role is the* rate $N^{-1/2}$ *and the uniformity over the candidate set, both matched empirically. It is finite exactly when* $m\varepsilon < \lambda_{\min}$ *and is reported as vacuous otherwise, never silently extrapolated.*

4 Detection dichotomy (background; demoted from the main line)

(a) If $S_T f_1 = S_T f_2$ and the observation law depends on the signal only through $S_T f$ (as for additive noise $\boldsymbol{y} = S_T f + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon}$ independent of f), the distributions of $\boldsymbol{y}$ under the two hypotheses are equal. For any (possibly randomized) test $\varphi(\boldsymbol{y}) \in [0, 1]$, $\mathbb{E}_{f_1}\varphi = \mathbb{E}_{f_2}\varphi$: power equals size.

(b) Under $\boldsymbol{y} = \boldsymbol{D}\boldsymbol{\beta} + \boldsymbol{s} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$ and $\boldsymbol{P}_D$ the projector onto the rank-p column space of $\boldsymbol{D}$, $(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{y} = (\boldsymbol{I} - \boldsymbol{P}_D)(\boldsymbol{s} + \boldsymbol{\varepsilon})$ is Gaussian with mean $(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{s}$ and covariance $\sigma^2(\boldsymbol{I} - \boldsymbol{P}_D)$; hence $\|(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{y}\|^2/\sigma^2$ is noncentral χ^2 with $N - p$ degrees of freedom and noncentrality $\lambda = \|(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{s}\|^2/\sigma^2$ (central when $\boldsymbol{s} = 0$ or when the component is coherent, $(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{s} = 0$). The stated power expression is the definition of the noncentral-χ^2 survival function at the central $(1-\alpha)$-quantile. $\square$

5 Background: average-case decomposition

Proposition S 2 (average-case LS error; standard bookkeeping)**.** *Let* $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{c}^\star + \boldsymbol{\Phi}_{\text{out}}\boldsymbol{a} + \boldsymbol{\varepsilon}$ *with* $\sigma_{\min}(\boldsymbol{\Phi}) > 0$*, the* a_l *zero-mean, uncorrelated,* $\mathbb{E}|a_l|^2 = p_l$*, independent of* $\boldsymbol{\varepsilon}$*, and* $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^*] = \sigma^2 \boldsymbol{I}$*. Then, with* $\boldsymbol{M} = (\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^*$,

$$\mathbb{E}\|\hat{\boldsymbol{c}} - \boldsymbol{c}^\star\|^2 = \sigma^2 \operatorname{tr}\big((\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}\big) + \operatorname{tr}\big(\boldsymbol{M}\,\boldsymbol{\Phi}_{\text{out}} \operatorname{diag}(\boldsymbol{p}) \boldsymbol{\Phi}_{\text{out}}^*\, \boldsymbol{M}^*\big).$$

Proof. $\hat{\boldsymbol{c}} - \boldsymbol{c}^\star = \boldsymbol{M}(\boldsymbol{\Phi}_{\text{out}}\boldsymbol{a} + \boldsymbol{\varepsilon})$; expand the squared norm; the cross term vanishes by independence and zero means; the two quadratic terms are the stated traces. $\square$

This proposition is retained for completeness only; it is standard omitted-variable bias/variance bookkeeping and is not claimed as a contribution.

6 Sampling design: surrogate and the continuum certificate

Relation to classical design. Write $\boldsymbol{\Psi} = \boldsymbol{\Phi}_\Omega$ for the atoms of the candidate out-of-band set Ω. The *exact* D_s-optimality (interest-vs-nuisance) criterion maximizes the log-determinant of the Schur complement $S = \frac{1}{N}(\boldsymbol{\Phi}^*\boldsymbol{\Phi} - \boldsymbol{\Phi}^*\boldsymbol{\Psi}(\boldsymbol{\Psi}^*\boldsymbol{\Psi})^{-1}\boldsymbol{\Psi}^*\boldsymbol{\Phi})$ of the interest block, treating Ω as a nuisance parameter block; S equals the isolated interest block iff the cross-block $\boldsymbol{\Phi}^*\boldsymbol{\Psi} = 0$, which is also the target of LCMV/MVDR null-steering. Our surrogate J (below) is *related to but not identical* to $\log\det S$—it penalizes the squared cross-block and the in-band off-diagonal directly—so we call it D_s/null-steering-*motivated*, credit the classical principle, and compare against the exact D_s design ("`ds_optimal`") as a baseline (§7). Our own additions are the identifiability reading (Theorem 1 says grid designs *cannot* realize the target for coherent folds) and the certificate below. The greedy/coordinate procedures are heuristics with deterministic outputs; **no approximation-ratio or submodularity guarantee is claimed**.

Proposition S 3 (surrogate controls aliasability)**.** *Let* $\mu(T) = \max_{k,\nu} |(\mathbf{\Phi}^*\mathbf{\Psi})_{k\nu}|/N$ *and let* $\lambda_0 = \lambda_{\min}(\mathbf{\Phi}^*\mathbf{\Phi}/N) > 0$*. Then for every* $\nu \in \Omega$*,* $a_T(\nu) = \|(\mathbf{\Phi}^*\mathbf{\Phi}/N)^{-1}(\mathbf{\Phi}^*\boldsymbol{\phi}_\nu/N)\| \leq \sqrt{m}\,\mu(T)/\lambda_0$*, and* $\mu(T)^2 \leq N^2 E_{\Lambda\Omega}(T)$ *where* $E_{\Lambda\Omega}$ *is the cross-energy term of the surrogate* J*. Hence minimizing* J *(which also drives* $\lambda_0 \to 1$ *through* $E_{\Lambda\Lambda}$*) upper-controls the worst-case aliasability over* Ω*.*

Proof. $\mathbf{\Phi}^*\boldsymbol{\phi}_\nu/N$ is the ν-column of $\mathbf{\Phi}^*\mathbf{\Psi}/N$, of norm $\leq \sqrt{m}\,\mu(T)$; multiply by $\|(\mathbf{\Phi}^*\mathbf{\Phi}/N)^{-1}\| = 1/\lambda_0$. The μ–E inequality is $\max \leq \ell_2$. No submodularity is claimed: a_T depends on $\mathbf{\Phi}^\dagger$ nonlinearly, so the greedy/coordinate procedures are heuristics with deterministic outputs, not approximation-ratio algorithms. □

Proposition S 4 (continuum certificate; states Proposition 1 of the main paper)**.** *Fix any design* T *with* $\sigma_{\min}(\mathbf{\Phi}) > 0$ *and let* $\boldsymbol{M} = \mathbf{\Phi}(\mathbf{\Phi}^*\mathbf{\Phi})^{-1}G_{L^2}(\mathbf{\Phi}^*\mathbf{\Phi})^{-1}\mathbf{\Phi}^*$ *(Hermitian PSD;* $G_{L^2} = \boldsymbol{I}$ *gives the coefficient norm). Then* $g(\nu) := a_{L^2,T}(\nu)^2 = \boldsymbol{\phi}_\nu^*\boldsymbol{M}\boldsymbol{\phi}_\nu = \sum_{j,l}\boldsymbol{M}_{jl}\,e^{i2\pi\nu(t_l - t_j)}$ *is a real* finite exponential polynomial *in* ν *(the exponents* $t_l - t_j$ *are generally non-integer, so this is a generalized, not periodic, trigonometric polynomial). Its derivative satisfies the explicit bound* $|g'(\nu)| \leq L := 2\pi\sum_{j,l}|\boldsymbol{M}_{jl}|\,|t_l - t_j|$*, so for any interval* $[a,b]$ *and any grid on it of spacing* h*,*

$$\max_{\nu\in[a,b]} a_{L^2,T}(\nu) \;\leq\; \Big(\max_{grid} g + L\,h/2\Big)^{1/2}.$$

The same argument applied to $h(\nu) = \boldsymbol{\phi}_\nu^*\boldsymbol{P}\boldsymbol{\phi}_\nu$ *(*$\boldsymbol{P} = \mathbf{\Phi}(\mathbf{\Phi}^*\mathbf{\Phi})^{-1}\mathbf{\Phi}^*$*) gives* $\min_{\nu\in[a,b]} v_T(\nu) \geq (1 - (\max_{grid} h + L_P h/2)/N)^{1/2}$*.*

Proof. $\Delta c = \mathbf{\Phi}^\dagger\boldsymbol{\phi}_\nu = (\mathbf{\Phi}^*\mathbf{\Phi})^{-1}\mathbf{\Phi}^*\boldsymbol{\phi}_\nu$, so $a_{L^2,T}(\nu)^2 = \Delta c^* G_{L^2}\Delta c = \boldsymbol{\phi}_\nu^*\boldsymbol{M}\boldsymbol{\phi}_\nu$ with $\boldsymbol{M}$ as stated (Hermitian since G_{L^2} is). With $\boldsymbol{\phi}_\nu[j] = e^{i2\pi\nu t_j}$, $g(\nu) = \sum_{j,l}\boldsymbol{M}_{jl}e^{i2\pi\nu(t_l - t_j)}$, real because $\boldsymbol{M}$ is Hermitian; then $g'(\nu) = \sum_{j,l}\boldsymbol{M}_{jl} i2\pi(t_l - t_j)e^{i2\pi\nu(t_l - t_j)}$ has $|g'| \leq 2\pi\sum_{j,l}|\boldsymbol{M}_{jl}||t_l - t_j| = L$ (triangle inequality; each exponential is unit-modulus—an elementary Lipschitz bound, a Bernstein-type inequality). Any point is within $h/2$ of a grid node, so $g(\nu) \leq \max_{\text{grid}} g + Lh/2$; take the sup and the (monotone) square root. The visibility bound is identical with $\boldsymbol{P}$. □

Everything in $\boldsymbol{M}, \boldsymbol{P}, L$ depends only on the sample times, so the certificate is a deterministic *post-hoc* guarantee applicable to *any* realized design (random, E-optimal, or optimized); it is what Theorem 2(b), a finite-candidate bound, does not by itself provide. Numerically it is computed via a Hermitian eigendecomposition of $\mathbf{\Phi}^*\mathbf{\Phi}$ (no explicit inverse) and reported as *vacuous* when $\mathbf{\Phi}$ is rank deficient (`full_rank=False`); soundness (certified $\geq$ true) and vacuity are tested in `tests/test_design.py` for random, near-singular, and rank-deficient designs.

7 Additional experimental results

These support the compressed statements in Sec. 6 of the paper. All figures are regenerated from the released `results/*.json` (seeds and environment stamped therein).

Detection (T4). Fig. 1. With calibration/test separation, we report five sample-only detectors (extended-dictionary ring test, in-sample residual, Lomb–Scargle band test, held-out prediction, cross-fit). *Only* the residual detector has a closed-form operating characteristic—the exact noncentral-χ^2 power curve of T4(b), which the empirical residual detector tracks; the other four are alternative empirical baselines shown for comparison and are *not* claimed to follow that curve. At the calibrated threshold the test-set false positive rate holds at its nominal 5%. The indistinguishability demonstration instantiates T4(a)'s two-point pair exactly—H1 places a tone at the grid-coherent ν=111, H0 the *same-amplitude* tone at its in-band twin

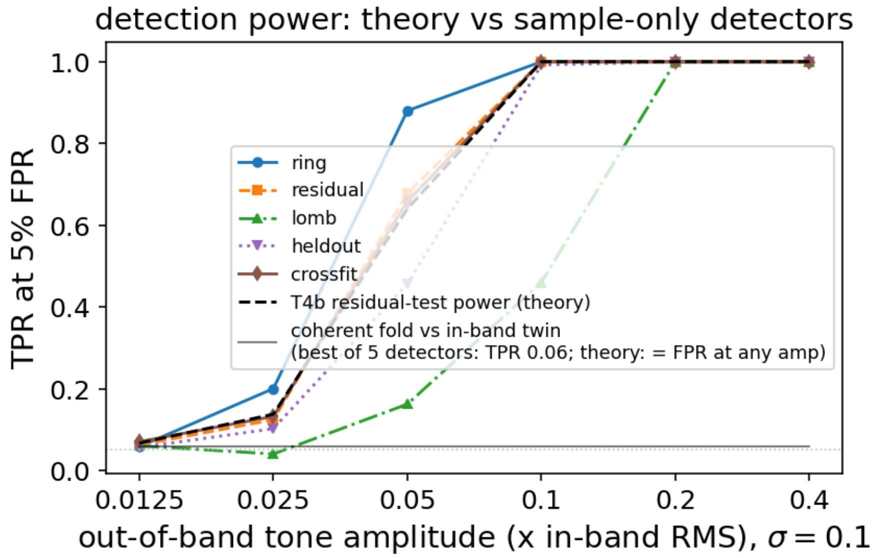

Figure 1: Detection power vs. tone amplitude; the coherent fold sits at chance.

$\nu - Q = -17$, so the grid sample laws are identical—and all five detectors sit at chance (best-of-five AUC 0.51, every bootstrap CI covering 0.5; best TPR at 5% FPR 0.06), at any amplitude. A no-tone null lets energy-sensitive statistics flag that *something* changed, but they cannot attribute it to out-of-band content; that attribution is what is impossible.

Trained-network extension (empirical only). Fig. 2. For Fourier-feature MLPs, SIRENs, and a band-limited head, tone attribution (an ablation: identical initialization trained with and without the tone) is compared to each network's own NTK-linearization prediction—no arbitrary reference band. Over ≥ 20 sampling scenarios—with feature and weight seeds drawn from independent streams (one triple per scenario, so they are decoupled from the sampling index rather than fully factorially crossed) plus label-permutation, amplitude-sweep-to-zero (infinitesimal-influence), and a weight-seed-stability control that varies only the weight seed at fixed sampling to isolate optimizer variance, with cluster-bootstrap CIs over sampling scenario—the linear picture extrapolates to *lazy-regime* FF-MLPs (exact fold match 75%, top-3 97%, pattern correlation median 0.96, n=60) and the band-limited head (median 0.86), but the trained-*SIREN* responses are *inconsistent with the initialization-NTK prediction* (exact match 30%, correlation median 0.47). We measured the init-vs-trained NTK Frobenius drift specifically to test whether this mismatch is a lazy-training / kernel-drift artifact, and it is *not*: the SIREN kernel drifts *less* than the FF-MLP kernel (median relative drift 0.64 vs. 6.92), yet the FF-MLP init-NTK prediction matches well while the SIREN one does not. The honest conclusion is therefore the negative one—the fixed-feature/init-NTK description does not extrapolate to trained SIRENs—and we explicitly do *not* attribute the failure to the SIREN "leaving" its initialization kernel. All seed-level failures are released.

Two dimensions. Fig. 3. The linear model reproduces the predicted 2-D frequency-vector fold exactly over a *held-out* tone set disjoint from the design dictionary (fold-exact rate 100%; e.g. planted $(27, 3)$ on a rate-32 grid subset folds to $(-5, 3)$, residual 10^{-15}). For trained 2-D networks (anti-aliased resize; images degraded with noise) the headline is the *excess* coherent-replica energy of lattice masks over matched-density *random* masks (the no-alias control), measured on the reconstruction-*error* spectrum so true image content does not inflate it: over ≥ 20 paired seeds on the synthetic image (sharp spectral lines) the excess is 1.791 and lattice exceeds random in 100% of seeds; the effect also replicates on the real *ascent* and *face* images in the per-configuration PSNR table (the ≥ 20-seed excess CI is reported only for the synthetic image); ridge and early stopping do not remove it—structured aliasing is a sampling-design property, not an overfitting artifact.

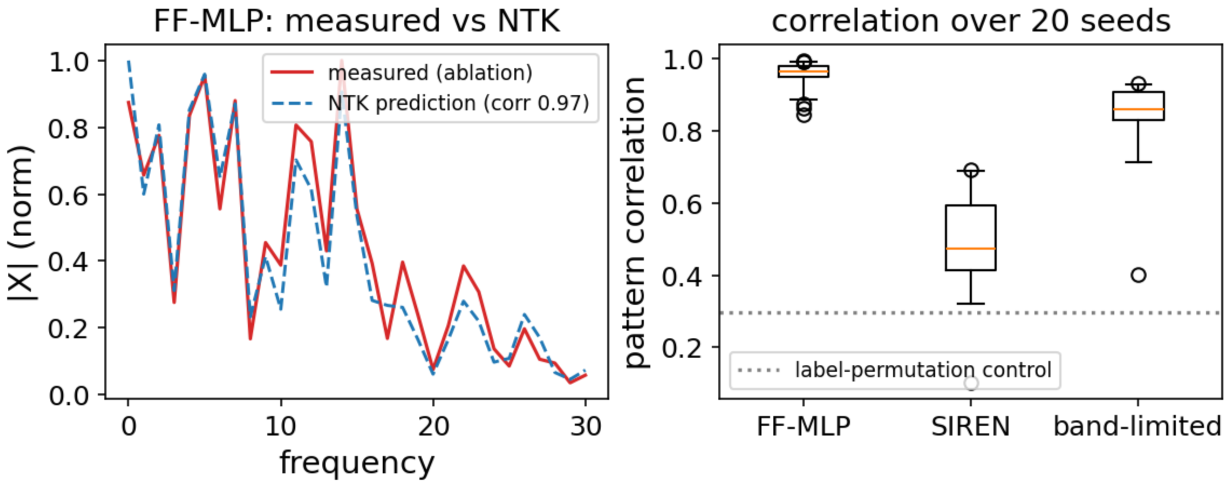

Figure 2: Trained-network tone attribution vs. NTK prediction; SIREN fails.

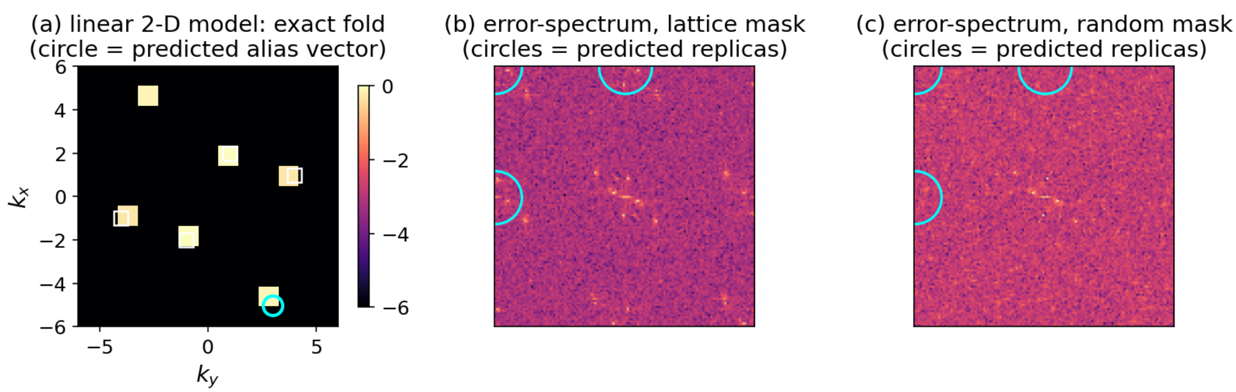

Figure 3: 2-D exact fold (linear) and lattice-vs-random replicas (trained).